

Formal Audit of Computational Complexity Barriers: A Graveyard of Failed P vs NP Proofs (2016-2026)

The determination of the relationship between the complexity classes P and NP represents the most significant unresolved challenge in theoretical computer science and mathematical logic. Since its formalization by Stephen Cook in 1971 and independently by Leonid Levin in 1973, the problem has transitioned from a niche inquiry in circuit complexity to a central pillar of the Millennium Prize problems.¹ As the mathematical community moves toward an era of computer-assisted orchestration—leveraging formal verification environments like Lean 4 and the inferential power of large-scale artificial intelligence—the landscape of the problem has shifted. We are no longer merely seeking a clever algorithm or a combinatorial trick; we are navigating a minefield of established "no-go" theorems and structural barriers that have claimed the efforts of thousands of researchers over the last decade. This report serves as a rigorous audit of the failed proof strategies and conceptual dead ends that have emerged between 2016 and 2026, specifically focusing on the intersection of Information Theory, Statistical Physics (FRSB), and Thermodynamic limits.

Foundations of the Computational Siege: P, NP, and the Proof Assistant Era

The class P encompasses decision problems solvable by a deterministic Turing machine in polynomial time, $O(n^k)$, while NP includes problems where a proposed solution can be verified in polynomial time.¹ The profound implications of the equality $P = NP$ were recognized early by visionaries like John Nash, who speculated in 1955 that the time required to crack complex codes increases exponentially with key length, and Kurt Gödel, who in 1956 asked John von Neumann whether theorem-proving could be automated.¹ The contemporary "siege" of this problem using Lean 4 is a response to a historical pattern of failure: most purported proofs contain subtle, informal gaps that human peer review often takes years to identify.⁵

Lean 4, as a dependent type theory-based proof assistant, provides an "antidote" to this human fallibility and the probabilistic unpredictability of modern AI systems.⁶ By requiring every inference step to be audited by a trusted kernel, Lean 4 ensures that if a proof of $P \neq NP$ compiles, the logical chain is sound—provided the axioms and definitions are not themselves degenerate.⁵ However, the graveyard of the last decade shows that even with formal tools, the primary failure mode is a conceptual mismatch between the physical laws of computation and

the abstract classes of complexity theory.

The Meta-Theoretic Dead Ends: Relativization, Natural Proofs, and Algebrization

Before addressing specific failures in Information Theory or Thermodynamics, any researcher orchestrating an AI-driven attack must confront the "meta-discoveries" that rule out vast categories of proof techniques. These are the three fundamental barriers that act as the gatekeepers of the P vs NP problem.⁸

The Relativization Barrier

In the 1970s, Baker, Gill, and Solovay demonstrated that there exist oracles A such that $P^A = NP^A$ and oracles B such that $P^B \neq NP^B$.⁸ This implies that any proof resolving P vs NP must use techniques that do not "relativize"—that is, they must exploit properties of computation that are specific to the standard Turing machine and fail when an arbitrary oracle is introduced.⁸ Most failed attempts between 2016 and 2026 that rely on diagonalization or simple simulation arguments fall into this trap, as they treat the machine's internal operations as a "black box".¹⁰

The Natural Proofs Barrier

Razborov and Rudich identified that most successful circuit lower bound techniques of the 1980s were "natural".⁸ A proof is natural if it identifies a property of Boolean functions that is "constructive" (decidable in $2^{O(n)}$ time), "large" (satisfied by a significant fraction of functions), and "useful" (proving functions in the class require large circuits).¹¹ If a natural proof were to separate P from NP, it would inadvertently prove that strong pseudorandom generators (PRGs) do not exist, a conclusion that contradicts the very cryptographic foundations upon which $P \neq NP$ is believed to rest.⁸ Any Lean 4 formalization that attempts to use combinatorial properties to bound circuit size must be audited for "largeness" and "constructivity" to ensure it does not collapse into a self-defeating natural proof.¹¹

The Algebrization Barrier

Aaronson and Wigderson extended these concepts in 2009 by showing that techniques based on "arithmetization"—representing Boolean formulas as low-degree polynomials over finite fields—are also insufficient.⁸ Arithmetization was successful in proving $IP = PSPACE$, but it cannot separate P from NP because such proofs hold even in "algebrized" worlds where the machine has access to a low-degree extension of an oracle.⁸ This barrier has rendered many "algebraic geometry" approaches to the problem, including early versions of Geometric

Complexity Theory (GCT), theoretically stagnant.⁸

Barrier	Description	Impact on Proof Orchestration
Relativization	Proof holds regardless of oracle access.	Rules out basic diagonalization and simulation.
Natural Proofs	Proof identifies a large, easy-to-check property.	Rules out most combinatorial circuit lower bounds.
Algebrization	Proof holds in algebraic oracle extensions.	Rules out pure arithmetization and simple algebraic geometry.

The Circuit Complexity Graveyard: Monotone and Non-Monotone Failures

The most visible failed proof of the last decade was the 2017 attempt by Norbert Blum, which claimed to prove $P \neq NP$ by leveraging monotone circuit complexity.¹³ Monotone circuits use only AND and OR gates, and it is well-known that the Clique problem requires exponential size in this model.¹⁵ Blum's strategy was to "lift" these monotone lower bounds to general circuits using "CNF-DNF approximators".¹³

The Norbert Blum Case Study (2017)

Blum's Theorem 6 asserted that if a monotone Boolean function has a certain type of approximator that proves a lower bound, then any general Boolean circuit computing that function must also be large.¹³ The technical failure of this proof was rooted in the "Tardos function," a specific monotone function that has small non-monotone circuits but requires huge monotone ones.¹³ Blum's argument failed because his method of approximating negated variables was found to be inconsistent with the behavior of these variables in standard Boolean networks.¹¹

Specifically, when an $\wedge$ -gate g has predecessors h_1 and h_2 , the reduced DNF

representation of g might lose information about which monomials are fulfilled by a specific input.¹³ This "bait and switch" problem meant that the inductive step of the proof, which tracked information gate-by-gate, could not account for how NOT gates could "unravel" the complexity at the very end of the computation.¹⁴ This highlights a critical "Dead End" for the Lean 4 graveyard: any attempt to generalize monotone results must explicitly account for the non-linear "compression" power of NOT gates.

Gate Elimination and Linear Bounds

Another dead end in the circuit complexity landscape is the reliance on "gate elimination" to prove linear lower bounds. Currently, the strongest known linear lower bound for general circuits is approximately $3.11n$ gates.¹⁶ The limitation of this method is its "local" nature—it relies on induction that eliminates gates based on fixed assignments, but it fails to capture the global structure of the circuit.¹⁶ Researchers have shown that these techniques are inherently capped and cannot reach the super-polynomial heights required to resolve $P \neq NP$.¹⁶

Geometric Complexity Theory (GCT) and the Multiplicity Obstruction Dead End

Geometric Complexity Theory, introduced by Mulmuley and Sohoni, was for years the "flagship" program for an algebraic attack on P vs NP.¹⁷ The strategy involved the "padded permanent" and comparing its orbit closure to that of the determinant.¹⁷ The program hoped to find "occurrence obstructions"—representation-theoretic multiplicities that were zero for the determinant but non-zero for the permanent, thereby proving they were in different orbit closures.¹⁷

The Ikenmeyer-Panova No-Go Theorems (2016-2022)

A major series of results by Christian Ikenmeyer and Greta Panova systematically dismantled the occurrence obstruction approach.¹⁷ They proved that for the rectangular Kronecker coefficients used in GCT, occurrence obstructions simply do not exist in the range required to prove super-polynomial lower bounds.¹⁹ Specifically, they showed that Kronecker coefficients $g(\lambda, \mu, \nu)$ grow at most polynomially for fixed Durfee square sizes, and that deciding the positivity of these coefficients is itself NP-hard.¹⁷

This created a "recursive trap": to use GCT to prove $P \neq NP$, one must first solve an NP-hard problem (computing Kronecker coefficients).²⁰ While "multiplicity obstructions" (where the multiplicities differ but are non-zero) might still exist, they are significantly harder to calculate and have not yielded any class separation.¹⁷ The "Dead End" here is the assumption that vanishing multiplicities in coordinate rings can act as a tractable proxy for computational

hardness.¹⁷

Information Theory and the Witness Incompressibility Mirage

The use of Kolmogorov complexity to separate P from NP is a strategy that has seen a resurgence in the last decade, notably in the 2025 Zenodo preprints by Andrew J. Murphy.²³ The core hypothesis is "Witness Incompressibility": the idea that NP-complete problems have a subset of instances whose witnesses are provably incompressible and possess high entropy that deterministic polynomial-time computation cannot reproduce.²³

The Murphy 2025 Failure

Murphy's proof claimed to establish a contradiction by showing that if $P = NP$, a uniform algorithm would exist that outputs incompressible witnesses from their inputs in polynomial time.²³ This would theoretically yield a program of description length $n + c$ that outputs a string of complexity exceeding $n + \delta$, a violation of Kolmogorov bounds.²³

The "Dead End" in this logic is the application of classical (time-unbounded) Kolmogorov complexity $K(x)$ to the time-bounded class P.²⁵ Kolmogorov himself believed that disproving the "chain rule" for time-bounded Kolmogorov complexity (K^t) was the path to proving $P \neq NP$, but no such refutation has been successfully formalized.²⁵ The failure of witness incompressibility proofs often stems from a "Semantic Identity Erosion": the algorithm for $P = NP$ does not need to *compress* the witness; it only needs to *find* it. The description length of the algorithm is a constant, and its execution on an input x to produce y does not necessarily violate incompressibility bounds because the "information" is already latent in the input-verifier relationship.¹²

Information Complexity and Search Problems

Research into "Interactive Information Complexity" has shown that solving certain problems, like Disjointness, requires a linear amount of information to be revealed.²⁸ However, translating these communication complexity bounds into circuit lower bounds remains an open challenge.²⁹ The "Dead End" is the belief that information density in the solution space directly correlates with algorithmic time complexity without accounting for the "lookup" or "derivation" cost that is not captured by entropy alone.³

Statistical Physics of Random CSPs: The FRSB and

Shattering Barriers

In the last ten years, the bridge between spin glass theory and computational complexity has revealed why random instances of NP-hard problems are difficult for local search algorithms.³⁰ This focuses on "Full Replica Symmetry Breaking" (FRSB) and the "shattering" of the solution space.³²

The Shattering Phase Transition

For random constraint satisfaction problems (CSPs) like k-SAT, as the ratio of constraints to variables (α) increases, the solution space undergoes a phase transition.³⁰ Initially, the space of solutions is a single connected component (Replica Symmetric). At a certain threshold, the space "shatters" into an exponential number of disjoint, isolated clusters.³² In the "Full Replica Symmetry Breaking" phase, these clusters become "frozen"—a subset of variables must take the same value in every solution within a cluster.³⁴

The "Dead End" for P vs NP proofs here is the assumption that the existence of these glassy phases in *random* instances implies *worst-case* hardness for all instances.³⁵ While shattering explains why algorithms like Belief Propagation fail, it does not provide a formal mechanism to separate P and NP globally, as a clever algorithm might still "tunnel" between clusters or exploit non-random structural regularities.³⁴ Furthermore, the "Contiguity Conjecture" suggests that these thresholds are optimal for local search, but not necessarily for global, non-local algorithms.³⁵

Phase Transition	Threshold Description	Algorithmic Barrier
Shattering	Solutions split into disjoint clusters.	Local search becomes trapped.
Condensation	A few clusters dominate the measure.	Stable sampling algorithms fail.
Freezing	Variables become fixed across clusters.	Algorithms cannot move between clusters.

SAT/UNSAT	Solutions cease to exist.	Decidability boundary for random CSPs.
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Thermodynamic Limits and the Landauer Principle

Dead End

A persistent trend in failed P vs NP proofs is the attempt to use the laws of thermodynamics to establish computational bounds.³ This approach typically hinges on Landauer's Principle: every logically irreversible bit operation (like erasure) requires a minimum energy dissipation of

$$E_{min} = k_B T \ln 2$$

The Reversibility Counter-Argument

The fundamental "Dead End" in thermodynamic proofs is the existence of "Reversible Computing".³⁷ Charles Bennett proved in 1973 that any deterministic computation can be performed reversibly with only a polynomial overhead in time and space.³⁹ Because reversible operations (like the Fredkin or Toffoli gates) do not erase information, they are not subject to the Landauer limit.³⁷

Therefore, any proof claiming that $P \neq NP$ due to energy dissipation requirements is mathematically invalid, as it only applies to *logically irreversible* implementations.³⁹ The energy cost of a computation is a function of the *hardware*, not the *complexity class*.³⁸ While "Physical NP Theory" explores resource-bounded complexity, it does not resolve the abstract P vs NP question, which is defined on a model with infinite time and zero energy constraints.³⁶

Thermal Profiles of NP Problems

Some research hypothesizes a "distinct thermodynamic profile" for NP problems compared to P problems, suggesting that search-intensive tasks have higher "computational entropy".³ However, the "Dead End" here is the lack of a formal mapping between physical entropy and algorithmic state transitions. High entropy might imply a "rugged solution landscape" in a physical system, but a Turing machine can navigate this landscape in a purely symbolic manner that bypasses thermal equilibrium assumptions.³

Refutations of the 2025 "Exhaustive Search" Claims

A particularly illustrative failure occurred in August 2025, when a paper titled "SAT Requires Exhaustive Search" was published in *Frontiers of Computer Science*.⁴² The authors, Xu et al., claimed that Constraint Satisfaction Problems (CSP) require time more than d^{cn} for any

constant $c < 1$.⁴²

The Williams/Allender Critique

Ryan Williams and Eric Allender provided a rigorous refutation, identifying two fatal flaws⁴²:

1. **Direct Mathematical Contradiction:** Williams had already presented an algorithm decades earlier that solves the $k = 2$ CSP case in $O(d^{0.8n})$ time.⁴² Since $0.8 < 1$, this directly refuted the paper's main theorem.⁴²
2. **Narrow Framework Fallacy:** The authors assumed that any algorithm for SAT must reduce an instance of size n to size $n - 1$ in a specific, known way.⁴² To prove $P \neq NP$, one must rule out every possible algorithm, including those not yet discovered and those that do not follow recursive reduction patterns.⁴²

This highlights the most common "Dead End" in modern proofs: the author unwittingly proves a lower bound for a *restricted class of algorithms* and then incorrectly claims it applies to the entire class P .⁴²

Formal Verification and AI Orchestration: Risks in the Lean 4 Siege

The user's intent to use Lean 4 to "orchestrate an AI siege" introduces a new layer of complexity. While Lean 4's type-checking ensures that the logic of a formalized proof is correct, several "Structural Audits" of the P vs NP framework in 2025-2026 have revealed significant pitfalls in formalization.⁵

The "Axiom Leakage" and "Sorry" Trap

In 2025, a machine-verified proof of $P \neq NP$ appeared on GitHub, containing over 15,000 lines of Lean 4 code.⁵ However, the proof relied on two explicit axiom declarations.⁵ Axioms in Lean are essentially logical "holes"—they represent statements that are assumed true without proof.⁵ The "Dead End" identified by the community was that these axioms, while seeming "tedious" to prove, actually contained the core complexity of the problem.⁵ Formal verification only proves that a conclusion follows from its axioms; it does not prove the axioms themselves are correct or consistent with the standard model of computation.⁵

Semantic Identity Erosion in AI-Driven Reductions

AI-driven formalization often fails at "Semantic Preservation".²⁷ When an AI agent performs a reduction (e.g., from SAT to Clique), it preserves the decision (YES/NO), but it may sever the "semantic identity" of the witness.²⁷ A structural audit in 2025 found that reductions often fail to

maintain:

- 1. **The Branching Factor:** The search space of the target problem might have a completely different topology than the original.²⁷
- 2. **Recursive Derivation:** The way a witness is derived in the reduction might not be "inversible" in polynomial time.²⁷
- 3. **State-Dependent Transitions:** The control flow of the verifier is often lost in the transformation.²⁷

If an AI-orchestrated proof relies on these transformations, it might prove a lower bound for a "lifted" version of the problem that has no bearing on the original P vs NP separation.¹⁰

The Graveyard: Condensed List of Dead Ends

For the purpose of the graveyard, the following table catalogs the technical reasons for failure in major categories over the last decade.

Dead End	Mathematical/Logical Basis of Failure	Reference
Monotone Approximation	Failure to account for non-monotone "compression" (Tardos Function).	Blum (2017) ¹³
GCT Occurrence Obstructions	Proven non-existence of obstructions for permanent vs determinant.	Ikenmeyer-Panova (2016) ¹⁷
Landauer Complexity	Reversible computing simulates irreversible computation with poly-overhead.	Bennett (1973/2003) ³⁹
Witness Incompressibility	Confusing uncomputable $K(x)$ with time-bounded algorithmic derivation.	Murphy (2025) ²³
Glassy/FRSB Hardness	Average-case clustering does not imply worst-case	Random CSP Research ³⁴

	hardness.	
Narrow Reductionism	Assuming SAT algorithms must follow known recursive patterns.	Williams/Allender Critique ⁴²
Axiomatic Formalization	Using Lean axioms to bypass combinatorial complexity of lower bounds.	GitHub Repo Controversy ⁵
Semantic Severing	Reductions that destroy witness topology and verifier control flow.	Structural Audit (2025) ²⁷

Strategic Guidance for the AI Siege

A successful proof of $P \neq NP$ using Lean 4 must navigate these failures by explicitly addressing the "Meta-Theory of Proof Complexity." The following third-order insights emerge from the graveyard:

The Convergence of Hardness and Randomness

The "Hardness vs Randomness" theory (Blum-Micali, Yao, Nisan-Wigderson) suggests that if $P \neq NP$ is true, then every efficient probabilistic algorithm can be made deterministic.² This implies that a separation proof might be more naturally framed as a "Derandomization" result. The "Dead End" to avoid is treating randomness as an *advantage* for NP that P cannot replicate; instead, one must prove that the *structure* required for efficient search is fundamentally different from the structure of verification.²

The Chain Rule for Time-Bounded Kolmogorov Complexity

The most promising, albeit currently failed, Information Theory approach involves the "Chain Rule for K^t ." It is known that $K^{2t}(x, y) \leq K^t(x) + K^t(y|x)$.²⁵ However, it is an open question whether a lower bound holds for some polynomial $p(t)$:
 $K^t(x, y) \geq K^{p(t)}(x) + K^{p(t)}(y|x)$.²⁵ Proving that this chain rule fails would be equivalent to $P \neq NP$.²⁵ Any Lean 4 orchestration should focus on formalizing the *failure* of this rule rather than just the incompressibility of witnesses.

The Limits of Autoformalization

AI agents in 2025-2026 struggle with "Catastrophic Forgetting" and "Sample Efficiency" when handling novel mathematical domains.⁴⁴ In the case of Isabelle/Lean formalization, AI has been shown to drop type annotations or lose "completeness-compromising" optimizations.⁴⁵ A human expert must formalize the "metatheory" on paper before letting an AI agent convert it to Lean code.⁴⁵ The "Dead End" is expecting an AI to "discover" a proof through pattern matching; the AI is merely a checker for the expert's structural intuition.⁴³

The Future Outlook: Stability, Safety, and the Aletheia Protocol

As we conclude this audit, the field is moving toward "Harmonic Computing" and the "Aletheia Protocol," which attempt to solve the P vs NP "paradox" by examining hardware topologies and universal constants.⁴⁶ While these are largely speculative, they point to a growing consensus:

$P \neq NP$ is likely a "structural" truth about the nature of information in our universe.⁴⁷ If

$P = NP$, the world would be "profoundly different"—there would be no "creative leaps," and recognizing a solution would be identical to finding it.⁴⁸

The graveyard of the last decade confirms that the "apparently hard problems" indeed have "inherent limits to algorithm efficiency".⁴⁹ For a researcher in 2026, the path forward in Lean 4 is not to seek a shorter path to a solution, but to rigorously formalize the *obstructions* themselves. The separation between P and NP is not just a gap in our knowledge; it is a chasm built into the logical structure of 3-SAT, resistant to relativization, natural proofs, and the mirages of thermodynamics.⁹ By formalizing these barriers as "impassable peaks and chasms," the Lean 4 siege can finally narrow the space in which $P = NP$ might still hide.²³

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